Foreign exchange

How do returns, expectations and purchasing power price a currency?

Today's route

Fix the quote and compare covered and uncovered returns.

Use purchasing power as a long-run benchmark.

Combine sticky prices and asset-market equilibrium to explain overshooting.

LECTURE 4

Currency quotes and contracts

What exactly are you buying when a currency strengthens?

One quote convention throughout

$$E_t = \text{home currency units per one foreign unit.}$$

E rises: the home currency depreciates.

E falls: the home currency appreciates.

A quotation example

Quote	Before	After
JPY per USD	150	135
JPY cost of USD 1,000	150,000	135,000

The yen appreciates. One dollar buys fewer yen.

Spot, forward and FX swap

Contract	Currency exchange dates
Spot	Near settlement at today's agreed rate
Forward	One future exchange at a rate agreed today
FX swap	An exchange now and a reverse exchange later

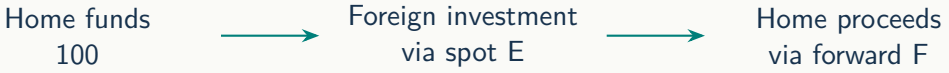
A forward FX rate is a contract price, not a promise about the later spot rate.

LECTURE 4

Covered interest parity

Can two locked currency routes pay different amounts?

Two routes for the same home-currency payoff



Compare this hedged route with lending directly at home.

The locked payoff equation

$$100(1 + R^{\text{home}}) = \frac{100}{E}(1 + R^{\text{foreign}})F.$$

$$\boxed{\frac{F}{E} = \frac{1 + R^{\text{home}}}{1 + R^{\text{foreign}}}.}$$

Covered interest parity, one matched period and consistent conventions.

R^{home}, R^{foreign}: matched-period rates. E: spot quote. F: forward quote.

A complete covered-parity calculation

$$E = 0.90 \text{ EUR per USD}, \quad R^{\mathrm{EUR}} = 2\%, \quad R^{\mathrm{USD}} = 5\%.$$

$$F = 0.90\frac{1.02}{1.05} = 0.874286 \text{ EUR per USD}.$$

$$100 \text{ EUR} \to 111.111 \text{ USD} \to 116.667 \text{ USD} \to 102 \text{ EUR}.$$

Why covered parity can show a basis

Dealer balance sheets, collateral and funding are costly.

The relevant borrowing rate can differ from the lending rate.

A quoted pricing gap need not be a scalable free profit.

LECTURE 4

Uncovered returns and carry

What changes when the future conversion rate is uncertain?

The unhedged foreign payoff

$$\text{Home-currency gross return} = (1 + R^{\text{foreign}})\frac{E_{t+1}}{E_t}.$$

E_{t+1} is the future spot quote. It is unknown today.

Uncovered interest parity is an equilibrium assumption

$$R^{\text{home}} \approx R^{\text{foreign}} + \mathbb{E}_t \Delta \ln E_{t+1}.$$

Equal expected returns require assumptions about risk and beliefs.

Currency risk premium

$$R^{\text{home}} - R^{\text{foreign}} - \mathbb{E}_t \Delta \ln E_{t+1} = \rho_t.$$

ρ_t: expected excess return required on the home asset.

The rate differential mixes expected FX change and compensation for risk.

Carry: borrow yen and invest dollars

$$E_t = 150 \text{ JPY per USD}, \quad R^{\text{JPY}} = 1\%, \quad R^{\text{USD}} = 5\%.$$

Borrow JPY 150 and buy USD 1.

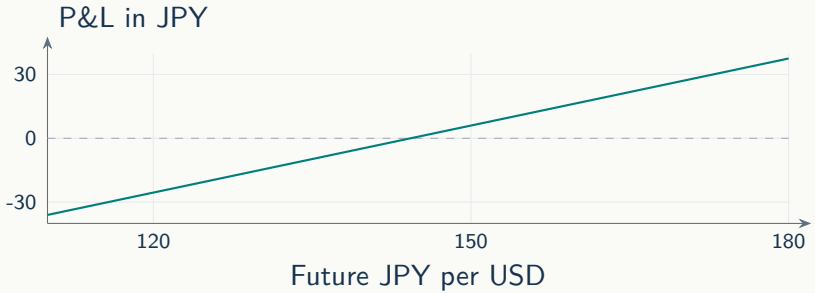
Next year receive USD 1.05 and owe JPY 151.50.

A small interest gain can meet a large FX loss

Future JPY per USD	JPY proceeds	Debt due	P&L
150	157.50	151.50	+6.00
135	141.75	151.50	-9.75

In the second scenario, JPY per USD falls 10%. Figures are hypothetical.

The carry payoff depends on the exit exchange rate



Break-even: 151.50 / 1.05 = 144.29 JPY per USD, before costs.

Crash risk can hide behind frequent small gains

Toy outcome	Probability	Normalised P&L
Quiet period	95%	+4%
Funding-currency surge	5%	-60%

$$0.95(4\%) + 0.05(-60\%) = 0.8\%.$$

An invented distribution, not an estimate of any currency strategy.

LECTURE 4

Expectations and purchasing power

Why can a rate increase mean appreciation now or weakness over time?

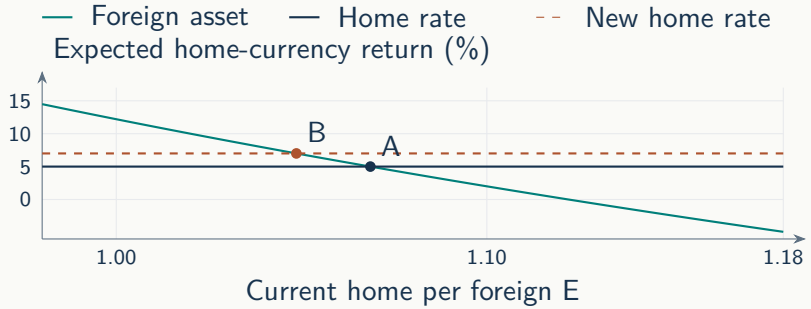
Short-run FX as an asset price

$$E_t = \frac{(1 + R^{\text{foreign}})E_t^e}{1 + R^{\text{home}}}.$$

E_t^e: expected next-period quote, fixed for this experiment.

A higher home rate then supports a stronger home currency.

A short-run FX return diagram



Expected future E = 1.10 and foreign rate 2%, both fixed. Hypothetical values.

The rate and expectation experiments

Change, other inputs fixed	Current home-per-foreign E
Home interest rate rises	Falls: home appreciates
Expected future E rises	Rises: home depreciates
Required home risk premium rises	Rises: home weakens

The third result uses the premium convention defined earlier.

Purchasing power parity

$$E^{PPP} = \frac{Q^{\text{home}}}{Q^{\text{foreign}}}.$$

A comparable basket costs 100 at home and 80 abroad.

PPP implies 1.25 home units per foreign unit.

Why a cheap haircut is hard to arbitrage

Haircuts require local labour and premises.

National baskets have different weights and taxes.

Productivity can change equilibrium relative prices.

Productivity and local service prices

Toy economy	Tradable productivity	Wage	Haircut cost
A	10 units / hour	10 / hour	10
B	5 units / hour	5 / hour	5

Tradable price 1 in both economies. One haircut per hour. Mobile domestic labour.

The real exchange rate

$$q = \frac{EQ^{\text{foreign}}}{Q^{\text{home}}}.$$

q: foreign-basket price relative to the home basket.

A rise in q is a real depreciation of the home currency.

Relative PPP and the Fisher effect

$$\Delta \ln E \approx \pi^{\text{home}} - \pi^{\text{foreign}}.$$

Persistent higher home inflation tends to erode the home currency.

Its higher nominal rate may simply compensate for that inflation.

Real interest differences and real FX expectations

$$R^{\text{real,home}} - R^{\text{real,foreign}} \approx \mathbb{E}_t \Delta \ln q + \rho_t.$$

Equal real rates need additional assumptions.

Expected real FX changes and risk premia can support persistent differences.

LECTURE 4

Dornbusch overshooting

Why can the currency jump further than its long-run adjustment?

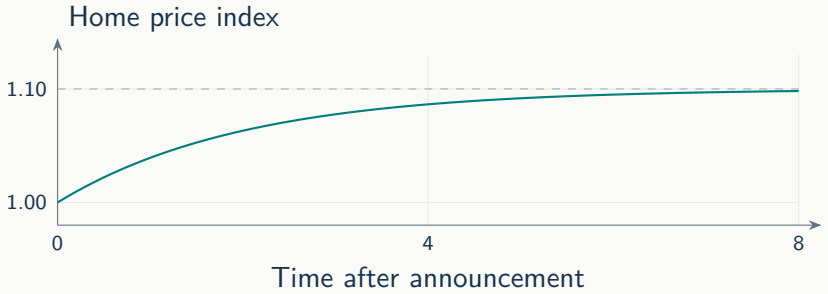
A permanent monetary level shock

The money stock rises permanently by 10% in the classic model.

Prices are sticky initially. Long-run output is fixed.

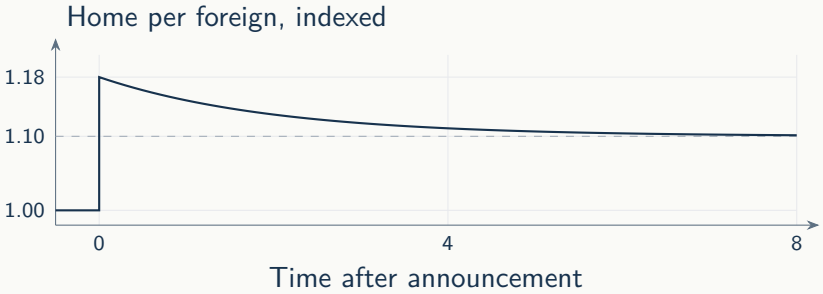
Flexible asset prices respond immediately.

Prices adjust slowly



Original illustrative transition. Initial level 1, long-run level 1.10.

The exchange rate can overshoot



The home currency weakens sharply, then appreciates toward its new long-run level.

The overshooting mechanism

Sticky prices
real money rises → Home rate falls
relative to foreign → Spot E jumps
then expected to fall

An expected future appreciation reconciles the lower home interest rate.

Exercise: three explanations for a high yield

Case A: an unexpected tightening, fixed future inflation and FX expectations.

Case B: higher expected persistent inflation.

Case C: a larger currency risk premium.

Further reading

Dmitriy Sergeyev, Bocconi 30054 course materials.

Krugman, Obstfeld & Melitz, *International Economics*.

Source links and worked answers accompany each slide in the notes.